

Notes: Itskhoki and Moll (Econometrica, 2019)

Optimal Development Policies with Financial Frictions

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1 Big picture

- **Question.** When entrepreneurs face borrowing constraints, is there a role for governments in developing countries to accelerate growth by intervening in labor, capital, and product markets? If yes, which policies are optimal over the transition to development?
- **Setting.** The paper studies **Ramsey-optimal dynamic policies** in a standard macro-development model with **heterogeneous entrepreneurs, workers, and financial frictions**. The baseline is a one-sector small open economy; the paper then adds a quantitative model, overlapping generations, household borrowing constraints, and a multi-sector extension.
- **Core challenge.** Financial frictions create misallocation because productive but poor entrepreneurs cannot scale up. Yet direct transfers to entrepreneurs may be infeasible or politically unrealistic, so the planner must decide whether *distortionary* product- and factor-market interventions can improve welfare and speed development.
- **Main theoretical result.** Optimal policy is **stage-dependent**.
 - Early in development, when entrepreneurs are undercapitalized, the planner adopts **pro-business** policies.
 - Later, once entrepreneurial wealth is sufficiently high, the planner switches to **pro-worker** policies.
- **Central mechanism.** Increasing labor supply early on suppresses equilibrium wages and raises entrepreneurial profits. Higher profits accelerate entrepreneurs' wealth accumulation, relax future collateral constraints, improve allocation and productivity, and eventually deliver **higher wages and lower work effort** for workers in the medium and long run.
- **Why it matters.** The paper provides an **efficiency rationale** for development policies often observed in fast-growing emerging economies—for example wage suppression, subsidized credit, and industrial targeting—while emphasizing an important caveat: these policies should **reverse** as the economy develops.
- **Main analytical results.**
 - In the baseline small open economy, the optimal policy does **not** distort workers' intertemporal saving margin, but it does distort labor supply.
 - The optimal labor wedge is **negative** when entrepreneurial wealth is scarce and **positive** in the long run.
 - This result survives the introduction of **credit subsidies, finite lives, worker borrowing constraints, and a present-biased planner**.
 - In the multi-sector extension, optimal policy subsidizes sectors with **latent comparative advantage**; under some instrument restrictions, it can also imply a **depreciated real exchange rate**.
- **Main quantitative results.**
 - Under a utilitarian planner, the optimal labor policy starts with about a **30% labor supply subsidy** and converges to a **long-run labor tax near 20%**.
 - The subsidy phase is persistent: its half-life is about **13 years**, and it takes about **17 years** before the subsidy becomes a tax.

- Relative to laissez-faire, GDP is almost **10% higher after 10 years** under the optimal policy.
- With an intermediate Pareto weight on entrepreneurs, the optimal labor tax raises overall welfare by about **0.5%** in consumption-equivalent terms and worker welfare by about **0.75%**; the optimal credit subsidy yields even larger gains.
- **Economic takeaway.** In financially underdeveloped economies, policies that look pro-business in the short run can be **worker-improving in present value**. The optimal policy mix is therefore dynamic: help entrepreneurs accumulate net worth early, then redistribute toward workers later.

2 Data and measurement (key objects)

2.1 Baseline economy and agents

- **Time and aggregate uncertainty.** The baseline model is in **continuous time**, has an **infinite horizon**, and features **no aggregate shocks**. The focus is entirely on transition dynamics.
- **Workers.** A representative worker household chooses consumption, labor supply, and bond holdings. Workers can save and borrow at the world interest rate in the baseline small open economy.
- **Entrepreneurs.** There is a unit mass of entrepreneurs. They differ in:
 - **wealth** a ,
 - **productivity** z .

Their joint distribution at time t is $G_t(a, z)$.

- **Political economy benchmark.** Much of the analysis sets the Pareto weight on entrepreneurs to **zero**, so any pro-business policy that emerges is not driven by favoritism toward capitalists.

2.2 Technology and financial frictions

- **Production.** Each entrepreneur operates a private technology that combines capital and labor to produce a homogeneous tradable good.
- **Collateral constraint.** Entrepreneurs face

$$k \leq \lambda a,$$

where $\lambda \geq 1$ measures financial development. The constraint ranges from self-financing at $\lambda = 1$ to frictionless capital markets as $\lambda \rightarrow \infty$.

- **Productivity process.** In the tractable baseline model, productivity shocks are i.i.d. over time and Pareto distributed with tail parameter $\eta > 1$.
- **Implication.** Productive but poor entrepreneurs cannot rent enough capital. As a result, capital is misallocated away from its most productive users.

2.3 Key aggregate objects and sufficient statistics

- **Aggregate entrepreneurial wealth:** $x(t)$.
- **Aggregate labor supply:** $\ell(t)$.
- **Aggregate capital demand:** $\kappa(t)$.
- **Aggregate output:** $y(t)$.
- **Aggregate entrepreneurial profits:** $\Pi(t)$.
- **Cutoff productivity for active entrepreneurs:** $\underline{z}(t)$.

These objects are enough to summarize transition dynamics in the analytical baseline.

- **Aggregate production takes a reduced form**

$$y = \Theta x^\gamma \ell^{1-\gamma},$$

where γ is the crucial sufficient statistic measuring how strongly entrepreneurial wealth matters for production.

- **Interpretation of γ .** The parameter γ is larger when:
 - production is more capital intensive,
 - entrepreneurial productivity is more heterogeneous,
 - financial frictions matter more for aggregate output.
- **Factor shares.** Labor still gets share $(1 - \alpha)$ of output, but entrepreneurs now earn positive profits. The presence of profits is a direct signal of financial frictions.

2.4 Inefficiency and policy instruments

- **Return wedge.** Workers earn the safe world return r^* , while constrained entrepreneurs earn a higher return because they are unable to scale up fully. This return wedge is the source of inefficiency.
- **Policy instruments in the baseline.**
 - labor income tax/subsidy $\tau_\ell(t)$,
 - savings tax/subsidy $\tau_b(t)$,
 - lump-sum transfer $T(t)$.
- **Additional instruments later in the paper.**
 - wage bill subsidy,
 - capital (credit) subsidy,
 - production/revenue/profit subsidies in extensions,
 - sector-specific labor and consumption taxes in the multi-sector model.

2.5 Quantitative model and calibration objects

- **Why a separate quantitative section is needed.** The tractable baseline relies on strong assumptions: constant returns, i.i.d. productivity, and unconstrained workers. The quantitative model relaxes these.
- **Main differences in the quantitative model.**
 - Production features **decreasing returns** with parameter $\beta < 1$.
 - Entrepreneur productivity follows a **persistent jump-diffusion** on a bounded support.
 - Workers are **hand-to-mouth** and cannot borrow.
 - Only a fraction ω of the population are entrepreneurs.
- **Initial condition.** The initial wealth distribution is taken to be a **ten-fold scaled-down version** of the laissez-faire stationary distribution. This makes the economy genuinely underdeveloped at the start of the transition.
- **Calibration highlights.**
 - $\lambda = 2$,
 - entrepreneur population share $\omega = 1/3$,
 - productivity persistence chosen to match an annual autocorrelation of about 0.85,
 - optimal policy is restricted to a simple parametric time path because the full dynamic state variable is the entire distribution $G_t(a, z)$.

3 Models and empirical specifications (all equations + interpretation)

3.1 Worker and entrepreneur problems

Worker preferences are

$$\int_0^\infty e^{-\rho t} u(c(t), \ell(t)) dt. \quad (1)$$

The worker's flow budget constraint is

$$c + \dot{b} \leq w\ell + r^*b. \quad (2)$$

Entrepreneurs have preferences

$$\mathbb{E}_0 \int_0^\infty e^{-\delta t} \log c^e(t) dt, \quad (3)$$

and produce with

$$A(zk)^\alpha n^{1-\alpha}. \quad (4)$$

They face the collateral constraint

$$k \leq \lambda a, \quad (5)$$

and wealth evolves according to

$$\dot{a} = \pi(a, z) + r^*a - c^e. \quad (6)$$

Per-period profit is

$$\pi(a, z) = \max_{n \geq 0, 0 \leq k \leq \lambda a} \{A(zk)^\alpha n^{1-\alpha} - wn - r^*k\}. \quad (7)$$

Interpretation.

- Workers are standard households supplying labor and holding bonds.
- Entrepreneurs differ from workers because they can operate productive technologies, but only up to the scale permitted by their own wealth.
- The planner is therefore dealing with a dynamic economy in which wealth distribution matters for efficiency.

3.2 Firm policy functions and the productivity cutoff

The solution to the entrepreneur's static problem implies

$$k(a, z) = \lambda a \cdot \mathbf{1}\{z \geq \underline{z}\}, \quad (8)$$

$$n(a, z) = \left(\frac{(1-\alpha)A}{w} \right)^{1/\alpha} z k(a, z), \quad (9)$$

and

$$\pi(a, z) = \left(\frac{z}{\underline{z}} - 1 \right) r^* k(a, z). \quad (10)$$

The cutoff productivity $\underline{z}$ satisfies

$$\alpha \left(\frac{1-\alpha}{w} \right)^{\frac{1-\alpha}{\alpha}} A^{1/\alpha} \underline{z} = r^*. \quad (11)$$

Under log utility, entrepreneur consumption is linear in wealth,

$$c^e = \delta a,$$

so wealth accumulation becomes

$$\dot{a} = \pi(a, z) + (r^* - \delta)a. \quad (12)$$

Interpretation.

- Only entrepreneurs with productivity above the cutoff $\underline{z}$ are active.
- Active entrepreneurs borrow up to the limit and operate at the collateral constraint.
- Profits are linear in wealth, which is the key tractability result that allows aggregation.

3.3 Aggregation and equilibrium

Aggregating across entrepreneurs yields

$$\kappa = \lambda x \underline{z}^{-\eta}, \quad (13)$$

$$\ell = \frac{\eta}{\eta - 1} \left(\frac{(1 - \alpha)A}{w} \right)^{1/\alpha} \lambda x \underline{z}^{1-\eta}. \quad (14)$$

Aggregate output can be written in the standard Cobb-Douglas form

$$y = Z \kappa^\alpha \ell^{1-\alpha}, \quad Z \equiv A \left(\frac{\eta}{\eta - 1} \underline{z} \right)^\alpha, \quad (15)$$

and also in the reduced-form wealth-labor form

$$y = \Theta x^\gamma \ell^{1-\gamma}, \quad \gamma \equiv \frac{\alpha/\eta}{\alpha/\eta + (1 - \alpha)}. \quad (16)$$

The cutoff productivity is pinned down by

$$\underline{z}^\eta = \frac{\eta \lambda}{\eta - 1} \frac{r^* x}{\alpha y}. \quad (17)$$

Income is split as

$$w\ell = (1 - \alpha)y, \quad r^* \kappa = \alpha \frac{\eta - 1}{\eta} y, \quad \Pi = \alpha \frac{1}{\eta} y. \quad (18)$$

Aggregate entrepreneurial wealth evolves according to

$$\dot{x} = \frac{\alpha}{\eta} y(x, \ell) + (r^* - \delta)x. \quad (19)$$

Interpretation.

- Entrepreneurial wealth x acts like an **accumulated productive factor**.
- Higher x raises output both directly and indirectly by allowing a higher productivity cutoff $\underline{z}$, meaning a more selective set of active entrepreneurs.
- Financial development and physical productivity enter aggregate output in very similar ways.

3.4 The inefficiency: a return wedge and a pecuniary externality

Workers earn the safe return r^* . By contrast, entrepreneurs with $z > \underline{z}$ earn a higher return on wealth. Averaging across entrepreneurs,

$$\mathbb{E}_z R(z) = r^* \left(1 + \frac{\lambda \underline{z}^{-\eta}}{\eta - 1} \right) = r^* + \frac{\alpha}{\eta} \frac{y}{x} > r^*. \quad (20)$$

Interpretation.

- The key inefficiency is a **return wedge**: resources in entrepreneurs' hands yield a higher return than resources in workers' hands.
- Workers do not internalize that supplying more labor today raises entrepreneurial profits, which speeds up wealth accumulation and increases future wages.
- This is a **dynamic pecuniary externality**, not a technological externality.

3.5 Policy environment and implementability

With labor and savings taxes, the worker budget constraint becomes

$$c + \dot{b} \leq (1 - \tau_\ell)w\ell + (r^* - \tau_b)b + T, \quad (21)$$

with balanced government budget

$$T = \tau_\ell w\ell + \tau_b b. \quad (22)$$

Household optimality implies

$$\frac{\dot{u}_c}{u_c} = \rho - r^* + \tau_b, \quad (23)$$

$$-\frac{u_\ell}{u_c} = (1 - \tau_\ell)w. \quad (24)$$

A crucial simplification is that any aggregate allocation satisfying

$$c + \dot{b} = (1 - \alpha)y(x, \ell) + r^*b, \quad (25)$$

$$\dot{x} = \frac{\alpha}{\eta}y(x, \ell) + (r^* - \delta)x, \quad (26)$$

can be decentralized by suitable taxes.

Interpretation.

- The planner can work directly with aggregate allocations rather than with all individual choices.
- This makes the Ramsey problem analytically tractable.

3.6 Optimality conditions and the stage-dependent labor tax

The planner chooses $\{c, \ell, b, x\}_{t \geq 0}$ to maximize worker welfare subject to the two implementability constraints above. The optimality conditions are

$$\frac{\dot{u}_c}{u_c} = \rho - r^* = 0, \quad (27)$$

$$-\frac{u_\ell}{u_c} = (1 - \gamma + \gamma\nu)(1 - \alpha)\frac{y}{\ell}, \quad (28)$$

$$\dot{\nu} = \delta\nu - (1 - \gamma + \gamma\nu)\frac{\alpha}{\eta}\frac{y}{x}. \quad (29)$$

The labor tax implied by these conditions is

$$\tau_\ell = \gamma(1 - \nu). \quad (30)$$

In steady state,

$$\bar{\tau}_\ell = \frac{\gamma}{\gamma + (1 - \gamma)(\delta/\rho)} > 0. \quad (31)$$

Interpretation.

- The planner never distorts the savings margin in the baseline small open economy: $\tau_b \equiv 0$.

- The sign of the labor wedge depends on ν , the shadow value of entrepreneurial wealth.
- If entrepreneurial wealth is scarce, then $\nu > 1$ and $\tau_\ell < 0$: the planner **subsidizes labor supply**.
- Once wealth is abundant enough, $\nu < 1$ and $\tau_\ell > 0$: the planner **taxes labor supply**.

There are two competing forces behind the labor-wedge formula above.

- **Static monopoly effect:** workers, taken collectively, restrict labor supply to raise wages; this pushes the planner toward a positive labor tax.
- **Dynamic productivity effect:** more labor today raises entrepreneurial profits and future productivity; this pushes the planner toward a labor subsidy.

The dynamic effect dominates early in development; the static redistributive effect dominates in the long run.

3.7 Additional instruments: wage bill and capital subsidies

Suppose the planner can also subsidize the wage bill and the cost of capital. Entrepreneur profits become

$$\pi(a, z) = \max_{n \geq 0, 0 \leq k \leq \lambda a} \{A(zk)^\alpha n^{1-\alpha} - (1 - \varsigma_w)wn - (1 - \varsigma_k)r^*k\}. \quad (33)$$

The optimal policy then uses both subsidies in tandem:

$$\frac{\varsigma_w}{1 - \varsigma_w} = \frac{\varsigma_k}{1 - \varsigma_k} = \frac{\alpha}{\eta}(\nu - 1). \quad (34)$$

Interpretation.

- Credit subsidies do *not* make labor subsidies redundant.
- A capital subsidy alone creates distortions of its own, especially on the extensive margin of entrepreneurial activity.
- The planner therefore combines labor and credit subsidies to support entrepreneurial net worth while minimizing new distortions.

3.8 Quantitative extension

The quantitative model replaces the tractable baseline with:

$$y = A[(zk)^\alpha n^{1-\alpha}]^\beta, \quad \beta < 1,$$

and entrepreneur productivity follows the jump-diffusion

$$d \log z_t = -\nu \log z_t dt + \sigma dW_t + dJ_t. \quad (37)$$

Because the state variable is now the whole distribution $G_t(a, z)$, the paper restricts the labor tax to the exponential path

$$\tau_\ell(t) = e^{-\gamma_\ell t} \tau_\ell + (1 - e^{-\gamma_\ell t}) \bar{\tau}_\ell. \quad (38)$$

The planner chooses policy parameters to maximize

$$V_0 = (1 - \omega) \int_0^\infty e^{-\rho t} u(c_t, \ell_t) dt + \theta \omega \int v_0(a, z) dG_0(a, z). \quad (39)$$

Interpretation.

- The quantitative section keeps the same economic mechanism but drops the assumptions that made the analytical solution possible.
- The paper then asks whether the stage-dependent logic survives in a much richer environment. It does.

3.9 Multi-sector extension

With multiple sectors, sectoral output satisfies

$$p_i y_i = p_i^\zeta \Theta_i x_i^\gamma \ell_i^{1-\gamma}, \quad \zeta \equiv 1 + \gamma(\eta - 1),$$

and sectoral wages are

$$w_i = (1 - \alpha) \frac{p_i y_i}{\ell_i}.$$

Sectoral labor allocation is

$$\ell_i = \frac{[(1 - \tau_i^\ell) p_i^\zeta \Theta_i x_i^\gamma]^{1/\gamma}}{\sum_{j=0}^n [(1 - \tau_j^\ell) p_j^\zeta \Theta_j x_j^\gamma]^{1/\gamma}} L. \quad (40)$$

For tradable sectors,

$$\tau_i^c = 0, \quad \tau_i^\ell = \gamma(1 - \nu_i), \quad i \in T. \quad (41)$$

For non-tradable sectors, the optimal policy imposes no overall sectoral wedge:

$$\tau_i = 0, \quad \tau_i^c = -\tau_i^\ell = \frac{1}{\eta - 1}(\nu_i - 1), \quad i \in N. \quad (42)$$

Interpretation.

- The planner tilts labor toward sectors with high shadow value of entrepreneurial wealth ν_i .
- In tradables, this looks like a labor subsidy to sectors with latent comparative advantage.
- In non-tradables, the least-distortionary way to raise profitability is often through consumption taxes offset by labor subsidies.

4 Identification and instruments (what makes the estimates credible)

4.1 What is inefficient in laissez-faire

- Laissez-faire is not just unequal; it is **dynamically inefficient**.

- The core inefficiency is that financially constrained entrepreneurs earn higher returns than workers, but private decisions do not move enough resources toward them.
- Because labor supply affects entrepreneurial profits and therefore future wealth accumulation, workers' labor choice creates a dynamic pecuniary externality.

4.2 What pins down the optimal policy

- The reduced-form aggregate production function

$$y = \Theta x^\gamma \ell^{1-\gamma}$$

makes the planner's problem depend on a small number of sufficient statistics.

- Two state-like objects are especially important:
 - x : entrepreneurial wealth,
 - ν : the shadow value of entrepreneurial wealth.
- The sign of the labor wedge is completely governed by ν :
 - $\nu > 1 \Rightarrow \tau_\ell < 0$,
 - $\nu < 1 \Rightarrow \tau_\ell > 0$.
- This gives the sharpest intuition in the paper: **the stage of development is summarized by the scarcity value of entrepreneurial wealth.**

4.3 Why direct credit tools do not eliminate the labor intervention

- One might think the planner should only use credit subsidies, since financial frictions are the underlying problem.
- The paper shows this is false: labor subsidies remain part of the optimum because direct credit support changes selection and creates additional distortions.
- Hence the optimal policy is not “credit only”; it is a **bundle** of pro-business instruments early on.

4.4 Why the results survive richer environments

- **Overlapping generations.** Finite worker lives do not overturn the logic. If the planner values all cohorts similarly, the analytical structure is essentially unchanged.
- **Present-biased planner.** Even when the planner discounts future generations more heavily, the key result survives: low entrepreneurial wealth still justifies early labor subsidies.
- **Worker borrowing constraints.** If workers are hand-to-mouth, the optimal labor subsidy becomes smaller in the short run, but the stage-dependent pattern remains.
- **Quantitative model.** After relaxing i.i.d. productivity, constant returns, and worker financial access, the quantitative results continue to favor pro-business policies early in development.

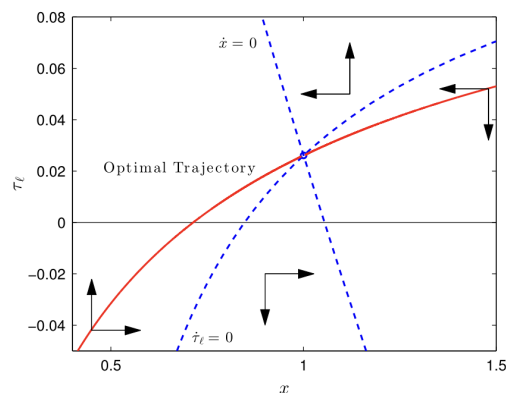


FIGURE 1.—Planner's allocation: phase diagram for transition dynamics.

4.5 Relation to the literature

- **Macro-development and finance.** The paper adds Ramsey policy analysis to a literature that studies how financial frictions depress TFP and slow development.
- **Ramsey taxation.** The method follows the dynamic public finance tradition, but the goal is different: the paper studies optimal intervention in a financially distorted growth economy rather than financing exogenous government spending.
- **Industrial policy.** The multi-sector extension links the model to classic debates on comparative advantage promotion, export-oriented growth, exchange-rate policy, and infant-industry protection.

5 Tables and figures

5.1 Figure 1: phase diagram for transition dynamics

Read:

- The horizontal axis is entrepreneurial wealth x and the vertical axis is the labor tax τ_ℓ .
- The optimal trajectory is the saddle path.
- Along that path, as wealth rises, the labor wedge increases: it starts **negative** and eventually becomes **positive**.
- The figure is the cleanest visual summary of the paper's main idea: **policy reverses sign as the economy develops**.

5.2 Figure 2 and Figure 3: planner allocation relative to laissez-faire

Read:

- Figure 2 shows that the planner initially sets a **labor subsidy**, then gradually turns it into a **labor tax**.

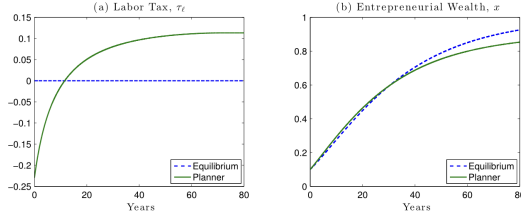


FIGURE 2.—Planner's allocation: labor tax $\tau_t(t)$ and entrepreneurial wealth $x(t)$. Note: In panel (b), the steady-state entrepreneurial wealth in the laissez-faire equilibrium is normalized to 1.

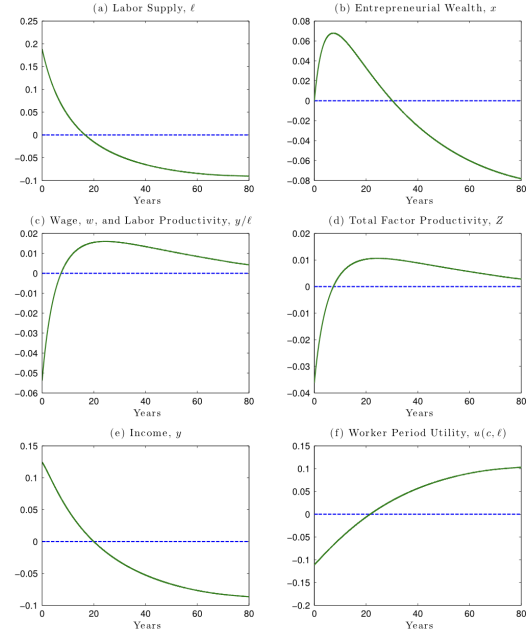


FIGURE 3.—Planner's allocation: proportional deviations from the laissez-faire equilibrium. Note: In panel (d), the deviations in TFP are the same as the deviations in z^* , as follows from (15). In panel (e), income deviations characterize simultaneously the deviations in output (y), wage bill ($w\ell$), profits (Π), capital income ($r^*\kappa$), and hence capital (κ), as follows from Lemma 2.

- Entrepreneurial wealth grows **faster** than under laissez-faire early on, because the subsidy raises profits and accumulation.
- Figure 3 shows the full transition:
 - labor supply is higher than laissez-faire at first, then lower later;
 - entrepreneurial wealth is above laissez-faire in the short run but below it in the long run because the planner eventually redistributes toward workers;
 - wages and labor productivity are initially lower, then become higher;
 - worker period utility is initially lower, then becomes higher.
- The key message is intertemporal: the planner is willing to hurt workers *today* to help them more *tomorrow*.

5.3 Figure 4: quantitative optimal policy and GDP paths

Read:

- The left panel plots the optimal labor tax schedules for different Pareto weights on entrepreneurs, plus the tax chosen by a myopic labor union.
- In all cases except the extreme worker-only benchmark, the optimal policy starts **more pro-business** than the long-run policy.
- Under the utilitarian planner, the policy starts with a large subsidy of about **30%** and converges to a tax near **20%**.
- The right panel shows that these policies raise GDP relative to laissez-faire during the transition, whereas the myopic labor-union policy depresses growth.

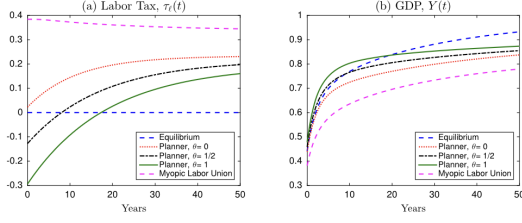


FIGURE 4.—Optimal policy in the quantitative model. Note: Panel (a) plots the optimal labor tax schedules $\tau_t(t)$ corresponding to different Pareto weights of entrepreneurs $\theta \in \{0, 1/2, 1\}$, as well as the labor tax imposed by a myopic labor union, as explained in the text. Panel (b) plots the evolution of GDP $Y(t) = \int y_t(a, z) dG_t(a, z)$, corresponding to the policies in panel (a), as well as the GDP dynamics under laissez-faire equilibrium, which is normalized to 1 in steady state.

TABLE I
WELFARE AND GROWTH EFFECTS OF POLICIES^a

	Welfare Gains (%)		Annualized GDP Growth (%)		
	Overall	Workers	5 Years	10 Years	20 Years
Optimal labor tax	0.503	0.751	9.59	5.51	3.01
Optimal flat labor tax, $\hat{\tau}_t = 0.100$	0.241	0.679	8.36	5.13	2.99
Flat labor subsidy, $-\hat{\tau}_t = 0.128$	-0.896	-1.442	10.16	6.05	3.46
Myopic union tax	-1.984	-0.378	5.38	3.65	2.28
Optimal credit subsidy	1.533	1.480	9.79	5.68	3.04

^aWelfare gains in consumption-equivalent terms (i.e., % increase in consumption in every period).

5.4 Table I: welfare and growth effects of alternative policies

Read:

- The **optimal labor tax** raises overall welfare by about **0.503%** and worker welfare by about **0.751%** in consumption-equivalent terms.
- The **optimal flat labor tax** performs noticeably worse than the stage-dependent tax, which shows that **timing matters**, not just the average size of the intervention.
- A permanent **flat labor subsidy** boosts growth early on but lowers welfare because the policy is not reversed.
- The **myopic labor union tax** is harmful for both growth and welfare.
- The **optimal credit subsidy** yields even larger welfare gains than the optimal labor tax, but it preserves the same stage-dependent logic: help entrepreneurs early, tax later.

5.5 Figure 5: two tradable sectors and latent comparative advantage

Read:

- Sector 0 has the latent long-run comparative advantage.
- The planner subsidizes labor toward that sector early on.
- As a consequence, sector-0 entrepreneurial wealth rises faster than under laissez-faire, while the less promising sector shrinks.

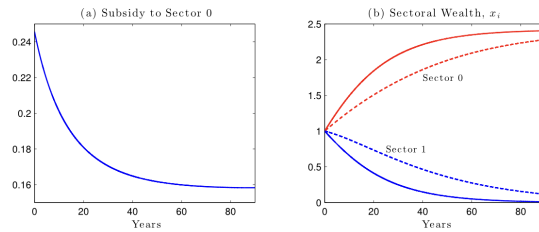


FIGURE 5.—Planner's allocation in an economy with two tradable sectors. Note: The sectors are symmetric in all but their latent comparative advantage, with $p_0^* \theta_0 > p_1^* \theta_1$. Panel (a) plots the labor supply subsidy to the comparative advantage sector 0. Panel (b) plots the evolution of the sectoral entrepreneurial wealth under laissez-faire (dashed lines) and optimal sectoral labor taxes (solid lines).

- The figure illustrates the paper’s industrial-policy result: **optimal policy tilts resources toward sectors with latent comparative advantage, not necessarily toward those that look strongest in the short run.**

6 Short summary

- **Method.** The paper solves a Ramsey problem in a growth model with heterogeneous entrepreneurs and collateral constraints, first analytically in a tractable one-sector environment and then quantitatively in a richer model.
- **Identification.** The planner’s problem collapses to a small set of aggregate state variables. Entrepreneurial wealth x and its shadow value ν summarize the key trade-off between redistribution today and productivity tomorrow.
- **Application / quantitative calibration.** The model is calibrated to a typical developing economy with severely undercapitalized entrepreneurs, persistent productivity heterogeneity, and financially constrained workers.
- **Main result.** Optimal development policy is stage-dependent: use pro-business policies such as wage suppression, labor subsidies, and credit support early on to accelerate entrepreneurial wealth accumulation; then reverse these policies later and tilt toward workers. In the multi-sector extension, the same logic implies targeted support for latent comparative advantage sectors.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- Financial frictions justify active development policy even in an otherwise standard competitive environment.
- The optimal policy is not constant over time. It starts pro-business and later becomes pro-worker.
- Policies that compress wages or subsidize production can therefore be efficient in early development, but only as **temporary** transition tools.

7.2 Economic intuition

- The planner wants more wealth in the hands of entrepreneurs when they are highly constrained, because that wealth relaxes future borrowing limits and raises aggregate productivity.
- Raising labor supply early is a way to transfer resources to entrepreneurs *indirectly*, through profits rather than through explicit lump-sum transfers.
- Once entrepreneurs have accumulated enough wealth, the productivity rationale weakens and redistributive motives dominate, so policy reverses.

7.3 Takeaway

This paper is important because it turns a broad idea—“some distortionary development policies may be good when finance is weak”—into a sharp dynamic Ramsey result. The deepest lesson is that **the right policy depends on the stage of development**. Early pro-business interventions can be welfare-improving, even for workers, but only if governments are willing to unwind them later.